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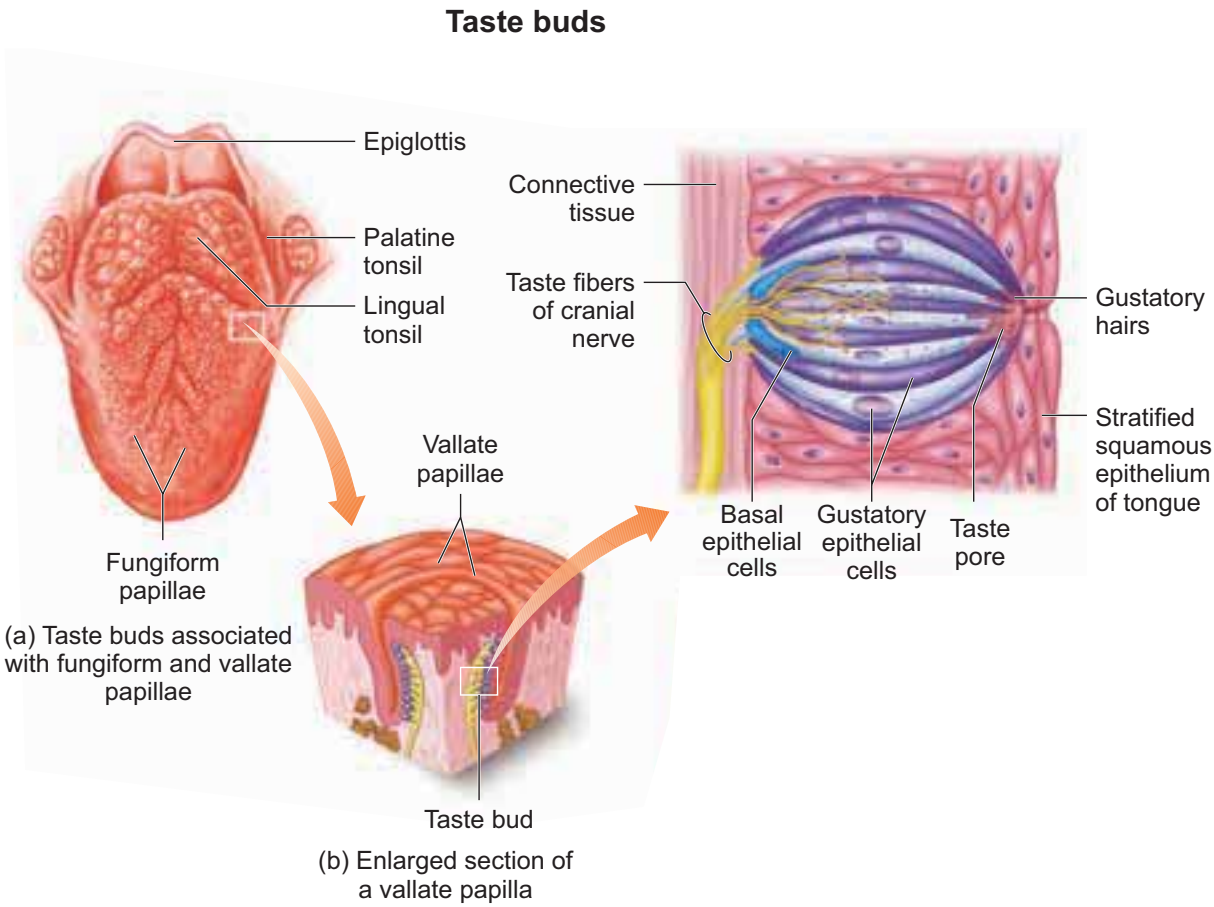


Figure 10.23 Gustatory receptor

Most taste buds are seen on the tongue (Figure 10.23) few are scattered on the soft palate, inner surface of the cheeks, pharynx and epiglottis of the larynx. Taste buds are flask-shaped and consist of 50 – 100 epithelial cells of two major types.

Gustatory epithelial cells (taste cells) and **Basal epithelial cells** (Repairing cells) Long microvilli called **gustatory hairs** project from the tip of the gustatory cells and extends through a taste pore to the surface of the epithelium where they are bathed by saliva. Gustatory hairs are the sensitive portion of the gustatory cells and they have sensory dendrites which send the signal to the brain. The basal cells that act as stem cells, divide and differentiate into new gustatory cells (Figure 10.23).

The taste bud cells are subjected to huge amounts of friction, because of their location and are routinely burned by hot foods. These are the most dynamic cells in the body and are replaced every seven to ten days.

Skin-Sense of touch

Skin is the sensory organ of touch and is also the largest sense organ. This sensation comes from millions of microscopic sensory receptors located all over the skin and associated with the general sensations of contact, pressure, heat, cold and pain. Some parts of the body, such as the finger tips have a large number of these receptors, making them more sensitive. Some of the sensory receptors present in the skin (Figure 10.24) are: